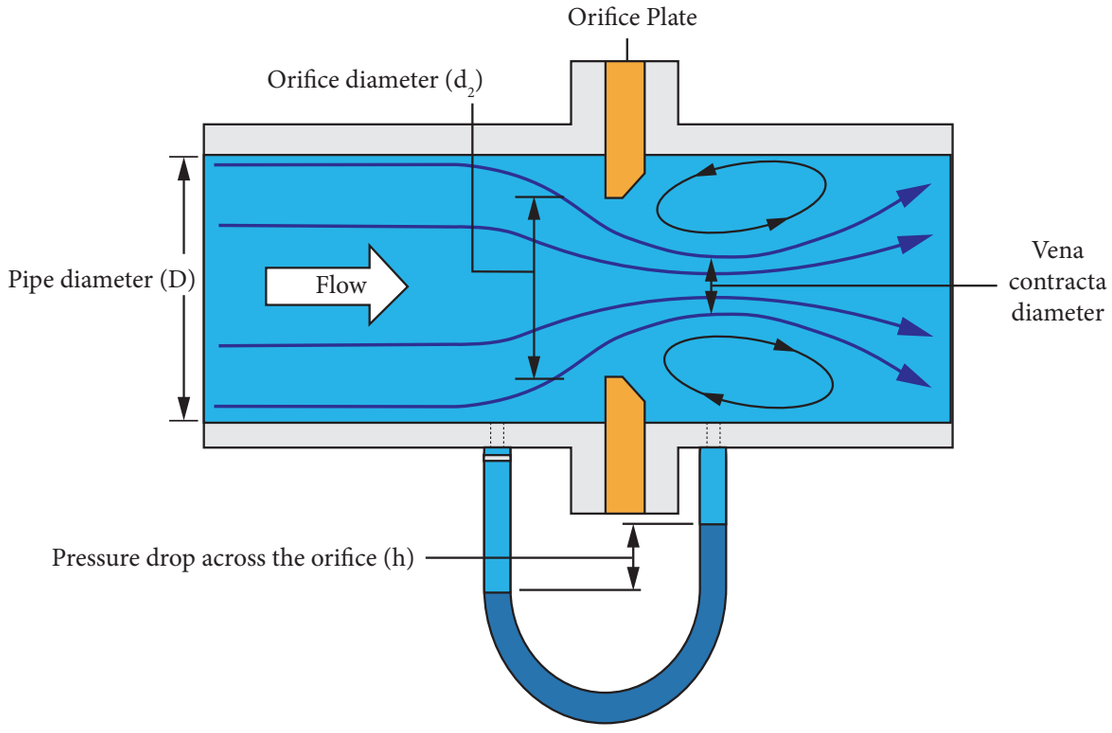


An Orifice plate with venacontracta

3. According to shape of the edge
   - a. Sharp-edged Orifice
   - b. Bell-Mouthed Orifice
4. According to nature of discharge
   - a. Fully submerged Orifice
   - b. Partially submerged Orifice

7.4.3 Jet of water

The continuous stream of water that comes out or flows out of an orifice / mouthpiece is known as jet of water.

7.4.4 Venacontracta

When a liquid flows out from a tank under pressure through an Orifice, the diameter of the jet of liquid will be equal to the diameter of the Orifice. But, the diameter of the jet and liquid contracts at a particular place away from the Orifice and then expands again. The point at which the diameter of the jet contracts to the maximum is called 'Venacontracta'.

7.4.5 Hydraulic Coefficients for small Orifice

The following three coefficients are known as hydraulic coefficients or Orifice Coefficients.

1. Coefficient of contraction, C_C
2. Coefficient of velocity, C_V
3. Coefficient of discharge, C_d.

1. Coefficient of contraction

The ratio of the area of jet at vena contracta to the area of the Orifice is known as coefficient of contraction. It is denoted by C_C.

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